

Resume

Aarif Khan

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Experience

Software Engineering Intern, Tekion Corp. — Bengaluru, India — Jan 2025–Present

- Completed a structured internship learning phase focused on Java, Spring Boot (AOP, transactions, REST), concurrency (ForkJoinPool, CompletableFuture), and design principles (SOLID, patterns), integrating Elasticsearch, Redis, and Kafka to build a performant, event-driven project with caching and search capabilities.
- Enabled forward compatibility in an existing Java microservice, ensuring smooth integration with future modules and enhancing system resilience.
- Designed and implemented a standalone application to separate concerns among services, improving codebase modularity and maintainability.

Software Developer Intern, Location Solutions, LLC — Dubai (Remote, Part-time) — Mar 2024–Aug 2024

- Rewrote a legacy project using Node.js, enhancing performance, scalability, and maintainability.
- Implemented RESTful APIs, resulting in a 20% performance increase and reduced response times.
- Implemented database join algorithms to execute in-memory joins on application data.
- Conducted comprehensive testing and debugging, and added unit testing to backend algorithms.
- Integrated OpenTelemetry logs into a legacy service to improve monitoring and troubleshooting capabilities.

Projects

Real Time Object Detection on a Video Stream from Drone/Mobile — Team/hackathon project, built during CodeUtsava 7.0 — [Source](#)

- Problem solved: pothole detection using drone footage and machine learning.
- Tools & technologies used: MERN stack, Flask, OpenCV, YOLOv8, RTMP protocol.
- Took an input stream from an RTMP source and ran object detection on the video frames.
- Built the detection script with OpenCV to run object detection and send frames to the server and database.
- Built an API to store detected frames into the file system and MongoDB database.
- Built a frontend UI in React to interact with the detected objects.

Video Conferencing Website — [Live](#) · [Source](#)

- Tools & technologies used: React.js, Express.js, Socket.io, WebRTC (Peer.js).
- Users create unique code-identified meeting rooms and share them to invite others.
- Built the backend server with Express.js to handle rooms and connection setup, using peer-to-peer WebRTC for video streaming and Socket.io for WebRTC signalling.

Multiplayer TicTacToe Game for Terminal — Playable over LAN — [Source](#)

- Tools & technologies used: Node.js, network sockets.
- Built the UI with terminal-kit to display game state on the terminal.
- Implemented a custom communication protocol over bare TCP.

Address Book (Website) — Stores wallet addresses with copy-to-clipboard access from any device

- Tools & technologies used: React.js, Express.js, MongoDB, JWT.
- Built a CRUD API on REST principles with Express.js, storing data in MongoDB.
- Implemented authentication using JWT.

Education

B.Tech in Computer Science and Engineering — National Institute of Technology, Raipur — 2025 CGPA/Percentage: 7.52

Class 12th (CBSE) — O. P. Jindal School, Raigarh — 2020 Percentage: 93.6%

Class 10th (CBSE) — O. P. Jindal School, Raigarh — 2018 Percentage: 91%

Skills

Languages: C++, JavaScript, Java

Developer Tools: VS Code, Git, Postman, Chrome Developer Tools

Frameworks: Spring Boot, Express.js, Node.js

Cloud / Databases: SQL, MongoDB, Kafka, Redis

Soft Skills: Communication, Team Collaboration, Problem-solving, Analytical Thinking

Areas of Interest: Operating Systems, Databases, Computer Networks

Achievements

- **Winner, CodeUtsava 7.0** — Central India's largest hackathon, organized by NIT Raipur — Nov 2023
- **Winner, CSE-MCA-IT branch, Vigyan Science Exhibition** — organized by NIT Raipur — Oct 2023
- **Codeforces Specialist** — Global Rank 530 in Codeforces Round 944 (Div. 4) — May 2024
- **Global Rank 2446/29648** in LeetCode Biweekly Contest 112
- **Solved 700+ DSA and problem-solving questions** across different platforms